Optimal Classification Tree for ER Triage (Extended Formulation)

1. Problem Statement

The objective is to learn a decision tree of maximum depth D that minimizes a combined cost function of classification error and feature measurement time, while strictly adhering to clinical workflow constraints (shallow triage vs. deep blood work) and remaining robust to bounded measurement noise in the patient's vital signs.

2. Sets and Indices

- $i \in \{1, \ldots, n\}$: Patients in the training dataset.
- $j \in \{1, \ldots, p\}$: Clinical features (e.g., HR, WBC).
- $k \in \{1, 2\}$: Classes (1 = Discharge, 2 = Admit).
- $t \in \mathcal{T}_B$: Set of branch nodes in the tree.
- $t \in \mathcal{T}_L$: Set of leaf nodes in the tree.
- $A_L(t), A_R(t)$: The set of left and right ancestor nodes for a given leaf t.

3. Parameters

- $X_{i,j} \in [0, 1]$: Normalized value of feature j for patient i.
- $Y_{i,k} \in \{0, 1\}$: 1 if patient i belongs to class k, 0 otherwise.
- ϵ_j: A sufficiently small constant to enforce strict inequalities for feature j.
- M_1, M_2, M_L: Big-M constants to activate/deactivate constraints.
- $N_{\min}$: Minimum number of patients required in a leaf node.
- τ_j: Time cost (in minutes) to measure feature j.
- α: Complexity parameter weighing feature time vs. accuracy.
- sev_w_i: Pre-computed severity weight for patient i based on baseline vital deviations.
- w_{FN}, w_{FP}: Base penalties for False Negatives and False Positives.
- E: Set of "expensive" features (where $\tau_j > 3.0$).
- $D_{\min}$: Minimum tree depth allowed for features in E.
- δ_j: Uncertainty budget (measurement noise bounds) for feature j.

4. Decision Variables

- $a_{j,t} \in \{0, 1\}$: 1 if branch node t splits on feature j.
- $b_t \in [0, 1]$: The continuous split threshold value at branch node t.
- $d_t \in \{0, 1\}$: 1 if branch node t applies any split at all (0 if pruned).
- $z_{i,t} \in \{0, 1\}$: 1 if patient i is routed to leaf node t.
- $l_t \in \{0, 1\}$: 1 if leaf node t contains at least one patient.
- $c_{k,t} \in \{0, 1\}$: 1 if leaf node t predicts class k.
- $FN_t \geq 0$: The total severity-weighted False Negative penalty at leaf t.
- $FP_t \geq 0$: The total False Positive count at leaf t.

5. Objective Function (Extension 1: Feature Time Costs)

Minimize the weighted misclassification cost plus the time cost of the features used:

$$\min \quad \frac{1}{n} \sum_{t \in \mathcal{T}_L} (w_{FN} FN_t + w_{FP} FP_t) + \alpha \sum_{t \in \mathcal{T}_B} \sum_{j=1}^{p} a_{j,t} \tau_j \tag{1}$$

Clinical Explanation: Standard AI assumes all information is available instantly. It might ask for a 2-hour blood test before checking a patient's temperature. To fix this, every test is assigned a "time penalty" (τ). The objective function forces the tree to prefer fast, cheap bedside vitals (like heart rate) over slow, expensive tests (like a lactate panel) unless the slow test is absolutely necessary for accuracy.

6. Constraints

A. Standard Tree Structure

$$\sum_{j=1}^{p} a_{j,t} = d_t \qquad \forall t \in \mathcal{T}_B \tag{2}$$

$$b_t \leq d_t \qquad \forall t \in \mathcal{T}_B \tag{3}$$

$$d_t \leq d_{p(t)} \qquad \forall t \in \mathcal{T}_B \setminus \{1\} \tag{4}$$

$$\sum_{t \in \mathcal{T}_L} z_{i,t} = 1 \qquad \forall i \tag{5}$$

$$\sum_{i=1}^{n} z_{i,t} \geq N_{\min} l_t \qquad \forall t \in \mathcal{T}_L \tag{6}$$

$$z_{i,t} \leq l_t \qquad \forall i, \forall t \in \mathcal{T}_L \tag{7}$$

$$\sum_{k=1}^{2} c_{k,t} = l_t \qquad \forall t \in \mathcal{T}_L \tag{8}$$

Mathematical Explanation: These are the foundational rules that make the decision tree function. They ensure that every branch node only splits on one feature (Eq. 2), that child nodes cannot split if their parent is pruned (Eq. 4), and that every patient is routed to exactly one active leaf node (Eq. 5, 7) that predicts exactly one class (Eq. 8).

B. Extension 2: Severity-Weighted Asymmetric Penalties

$$FN_t \geq \sum_{i=1}^{n} (sev\_w_i \cdot Y_{i,2} \cdot z_{i,t}) - M_L(1 - c_{1,t}) \qquad \forall t \in \mathcal{T}_L \tag{9}$$

$$FP_t \geq \sum_{i=1}^{n} (Y_{i,1} \cdot z_{i,t}) - M_L(1 - c_{2,t}) \qquad \forall t \in \mathcal{T}_L \tag{10}$$

Clinical Explanation: Standard AI minimizes overall error, treating a False Negative (discharging a critical patient) exactly the same as a False Positive (admitting a healthy patient for observation). Here, the model heavily penalizes False Negatives (w_{FN}). Furthermore, a severity score (sev_w_i) dynamically scales this penalty based on how abnormal the patient's baseline vitals are. This forces the tree to be highly conservative, prioritizing human life over statistical accuracy.

C. Extension 3: Depth Constraints

$$a_{j,t} = 0 \qquad \forall j \in E, \ \forall t \in \mathcal{T}_B \text{ where } \mathrm{depth}(t) < D_{\min} \tag{11}$$

Clinical Explanation: Even with time penalties, optimization anomalies can place an expensive blood test at the very root of the tree, violating clinical triage sequences. This hard constraint mathematically forbids the AI from using an expensive test for the first few splits, ensuring the diagnostic pathway always begins with basic bedside vitals.

D. Extension 4: Sensitivity Analysis under Measurement Uncertainty

In LP duality, the shadow price p_i^* of a constraint measures the rate of change of the optimal objective when the right-hand side b_i is perturbed:

$$p_i^* = \lim_{t \to 0} \frac{v(b + t \cdot e_i) - v(b)}{t} \tag{12}$$

In our OCT formulation, the split threshold b_m at each branch node is the RHS of the routing constraints. A small perturbation to b_m — caused by measurement noise $\delta_{j(m)}$ — shifts which patients satisfy the routing condition, changing the misclassification cost. The sensitivity of node m to this perturbation is therefore determined by how many patients sit within the uncertainty band $\pm\delta_{j(m)}$ of the threshold.

We estimate this as a severity-weighted at-risk count:

$$p_m^* \approx \frac{1}{n} \sum_{i=1}^{n} sev\_w_i \cdot \mathbf{1}\left[|x_{i,j(m)} - b_m| \leq \delta_{j(m)}\right] \tag{13}$$

where sev_w_i is the severity weight from Extension 2, prioritising critically ill borderline patients. The adjusted sensitivity is then:

$$\tilde{S}_m = p_m^* \times \delta_{j(m)} \tag{14}$$

This jointly captures the density of borderline patients near the threshold and the inherent unreliability of that measurement — the two factors that determine how much the optimal cost degrades under realistic noise. Splits are classified as **STABLE** ($\tilde{S}_m < 0.05$), **MODERATE** ($0.05 \leq \tilde{S}_m < 0.15$), or **FRAGILE** ($\tilde{S}_m \geq 0.15$).

Results: Pain Score < 3.73 was identified as **FRAGILE** ($\tilde{S}_m = 0.239$, 68 at-risk patients, $\delta = 1.0$), consistent with its high subjectivity as a self-reported measure. GCS splits were **MODERATE** due to inter-rater variability, while Temperature and Lactate were **STABLE** owing to high instrument precision. These results directly inform clinical practice — a **FRAGILE** split threshold should be double-checked before acting on the tree's decision.